Lecture 2: Validation, Overfitting, and Forecast Failure

ECON 417 Business Forecasting

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How Lecture 2 Fits the Course

- Lecture 1: we learned **how to build a baseline forecast**.
- Lecture 2: we learn **when not to trust a forecast**.
- Lecture 3: we will learn **how to stabilize unstable models**.

Key message: Good forecasting is not only fitting. It is controlled skepticism.

Lecture Map

1. Why Validation Matters
2. Overfitting vs Underfitting
3. Time-Aware Validation Design
4. Rolling and Expanding Windows
5. Cross-Validation with Time-Series Caveats
6. Bootstrap and Forecast Uncertainty
7. Why Forecasts Fail
8. Wrap-up

Warm-up Question

Suppose two models are both trained on the same history:

- Model A: very low in-sample error.
- Model B: slightly higher in-sample error.

Question: Which one should we deploy next quarter?

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Answer

We cannot decide from training fit alone. We need out-of-sample validation.

Big Idea of Validation

- A model can look brilliant on old data and fail immediately on new data.
- Validation asks: **Can this model survive new periods?**
- In forecasting, validation must respect the time arrow:

$$\text{Past} \rightarrow \text{Present} \rightarrow \text{Future}$$

Key message: If your evaluation design is wrong, your model choice is wrong.

Intuition First

Overfitting

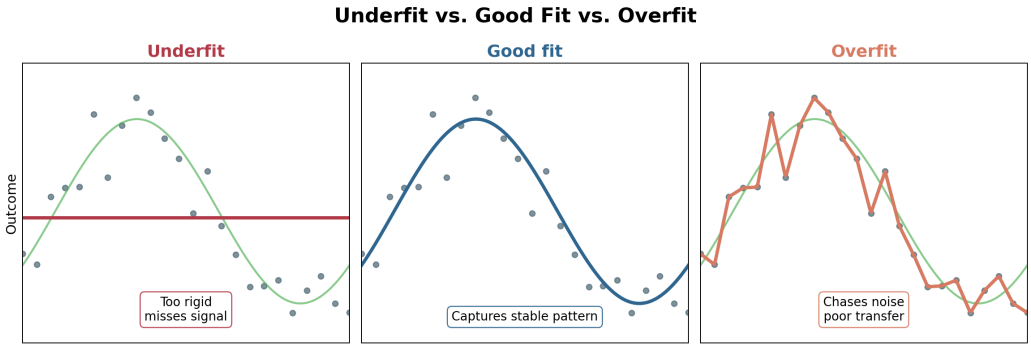
- Memorizes noise.
- Looks great in training.
- Performance drops on new data.

Underfitting

- Too rigid.
- Misses real signal.
- Both train and test errors are high.

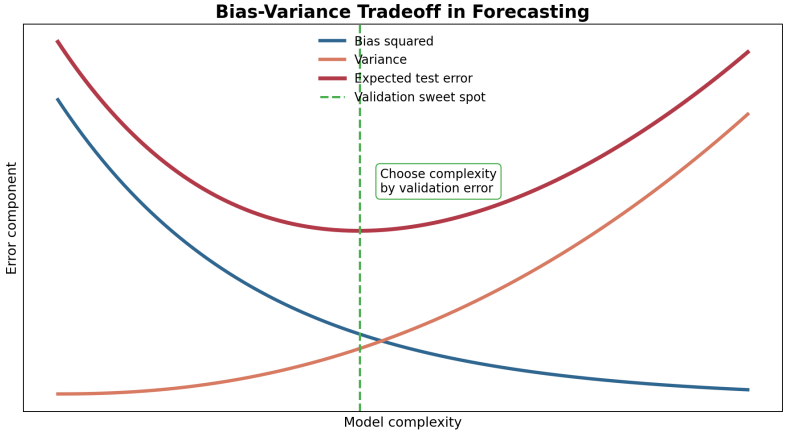
Key message: Our goal is not minimum training error; it is reliable future error.

Underfit vs Good Fit vs Overfit



graph from: IBM

Bias-Variance Tradeoff



graph from: IBM

Fit vs Generalization Summary

- **Underfitting**: model is too simple, so it misses real signal.
- **Overfitting**: model is too tailored to training data, including noise.
- **Good fit**: model captures stable pattern and transfers to unseen data.
- **Practical takeaway**: evaluate on future-like data, not only training fit.

Key message: Generalization is the target; training perfection is not.

Diagnostic with RMSE

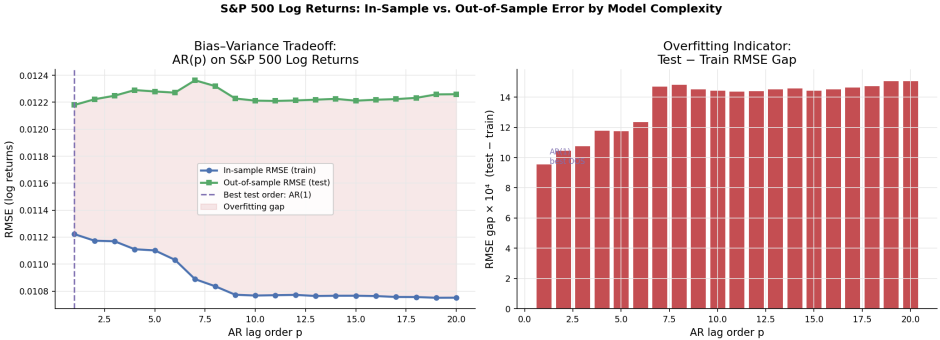
$$\text{RMSE}_{\text{train}} = \sqrt{\frac{1}{n_{tr}} \sum (y_t - \hat{y}_t)^2}$$

$$\text{RMSE}_{\text{test}} = \sqrt{\frac{1}{n_{te}} \sum (y_t - \hat{y}_t)^2}$$

- If complexity increases and $\text{RMSE}_{\text{train}}$ keeps falling, this is normal.
- If $\text{RMSE}_{\text{test}}$ starts rising, overfitting begins.
- Useful warning sign:

$$\text{Gap} = \text{RMSE}_{\text{test}} - \text{RMSE}_{\text{train}} \uparrow$$

Notebook Example: AR(p) on S&P 500 Returns



- As p grows, train error usually improves.
- Test error improves first, then worsens.

Key message: More parameters can mean less forecasting power.

Everyday Analogy

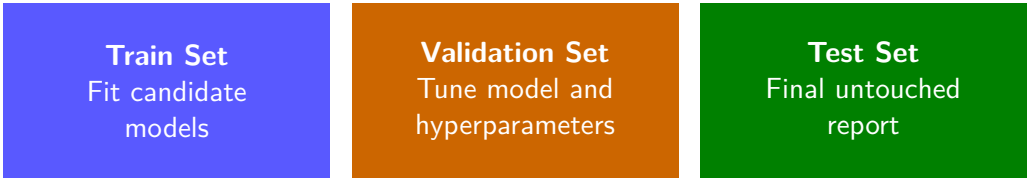
Student analogy

Memorizing past exam answers $\neq$ understanding the course.

- Overfitting model = student who memorizes specific questions.
- Robust model = student who understands core logic.

Class prompt: What kind of model would you hire for an unknown future?

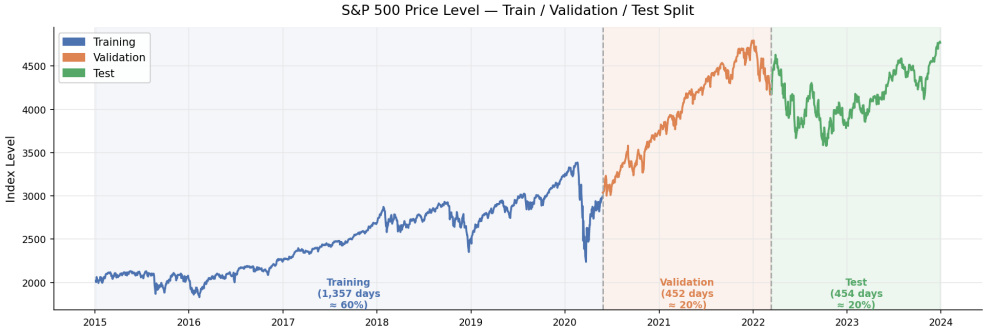
Train / Validation / Test Split



- Datasets are typically split into train, validation, and test sets.
- Use this sequence to estimate, tune, and then report a final untouched result.

graph adapted from: IBM

Notebook Example: Chronological Split



- Train set: estimate candidate models.
- Validation set: choose model/hyperparameters.
- Test set: untouched final report.

The Gold Rule: Protect the Test Set

- Do not repeatedly tune based on test performance.
- If test data guides your choices, it becomes another validation set.
- Then your "final" score is optimistic.

Practical rule

Touch test set once, at the end.

Data Leakage: Why It Is Dangerous

- Leakage means future information sneaks into training.
- Result: model appears powerful but cannot be deployed honestly.

One invalid example

Using y_{t+1} to predict y_t in real-time forecasting.

Key message: If you could not know it at forecast time, you cannot use it.

Data Leakage Types

Target Leakage
Uses future-only information not available at prediction time

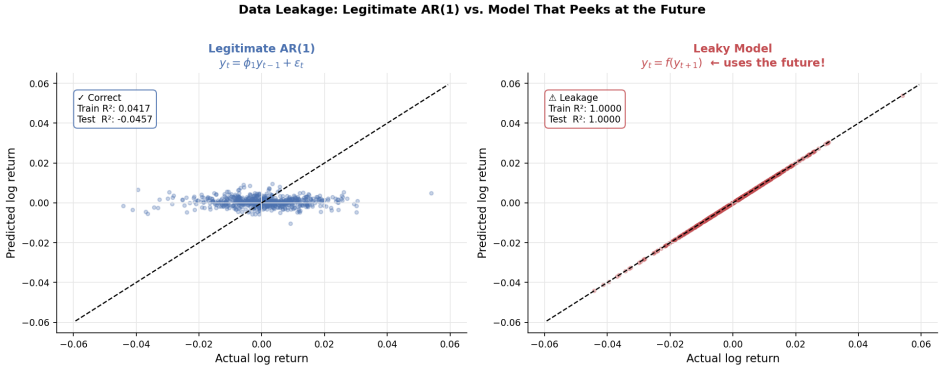
Train-Test Contamination
Test information leaks through preprocessing or split design

Leakage $\Rightarrow$ Inflated validation score $\Rightarrow$ Deployment failure

- Target leakage and train-test contamination are two key failure patterns.
- Both make a model look better during development than in real forecasting use.

concept adapted from: IBM

Notebook Example: Legitimate vs Leaky Model



- Leaky specification gives misleadingly strong fit.
- Real forecasting performance will collapse.

Two Repeated Validation Schemes

Rolling window

- Fixed history length.
- Focuses on recent regime.
- Faster adaptation.

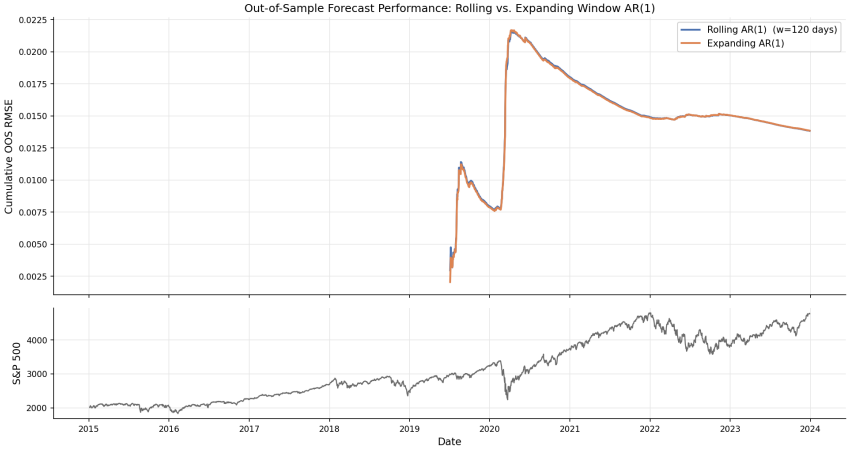
Expanding window

- History keeps growing.
- Lower variance.
- Better if process is stable.

Window Mechanics (Visual)



Empirical Comparison from Notebook



- No method dominates in all periods.
- Method choice depends on instability level.

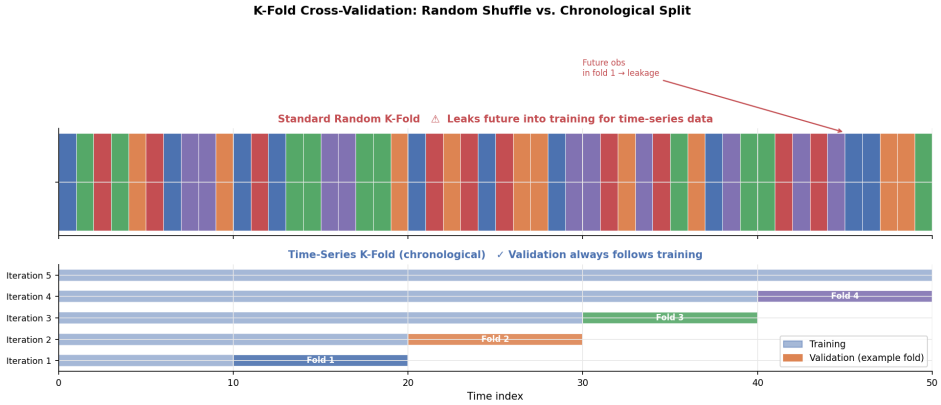
Why Random K-fold Is Risky Here

- Random K-fold assumes observations are exchangeable.
- Time series are ordered and dependent.
- Random shuffling can create hidden look-ahead bias.

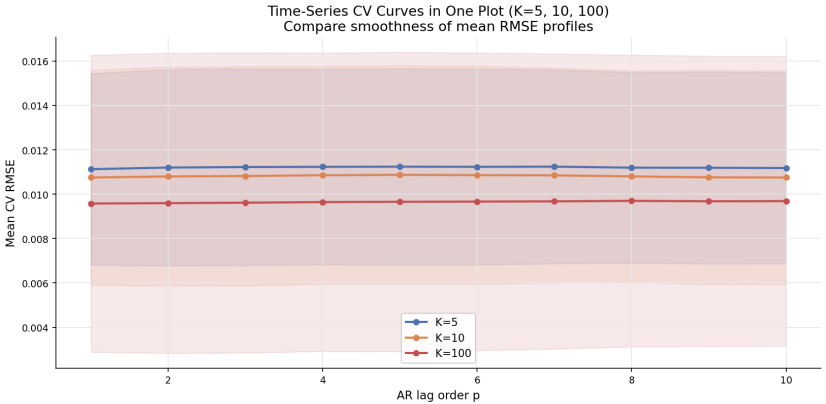
Safe principle

Each validation fold must be later in time than its training fold.

Standard K-fold vs Time-Series K-fold



Fold Count Tradeoff (K=5,10,100)



- More folds can smooth averages but may increase compute and variance sensitivity.

What Is Bootstrapping?

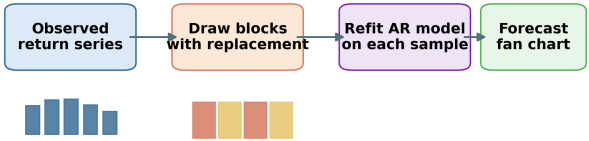
Bootstrap definition

Estimate uncertainty by repeatedly drawing samples with replacement, recomputing the statistic, and using those values to approximate its sampling distribution.

- Keep sample size fixed.
- Repeat many times (B large).
- Use the empirical distribution for SE/intervals.

definition adapted from: NIST/SEMATECH e-Handbook

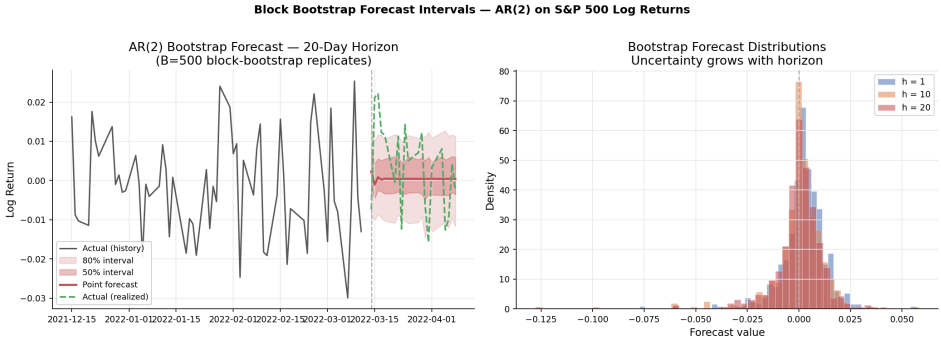
Block Bootstrap Preserves Local Time Dependence



Why We Need Intervals, Not Just Points

- Decision-makers need uncertainty bounds, not one number.
- Block bootstrap preserves local time dependence.
- Output: predictive fan chart across forecast horizons.

Notebook Example: Block Bootstrap Intervals



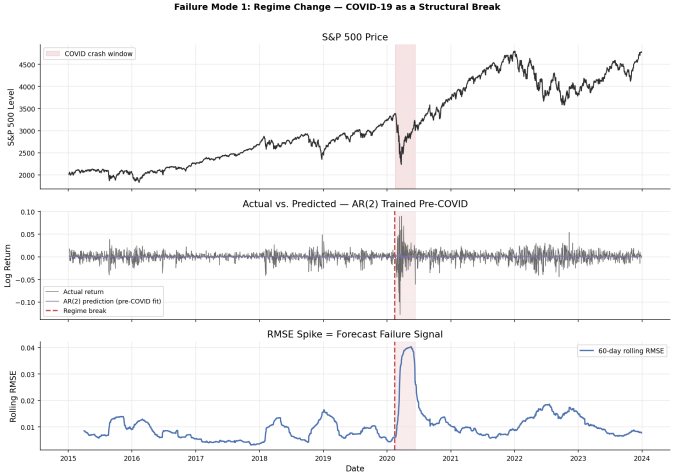
- Bands widen with horizon.
- Wider bands reflect uncertainty accumulating with the forecast horizon.

Three Common Failure Modes

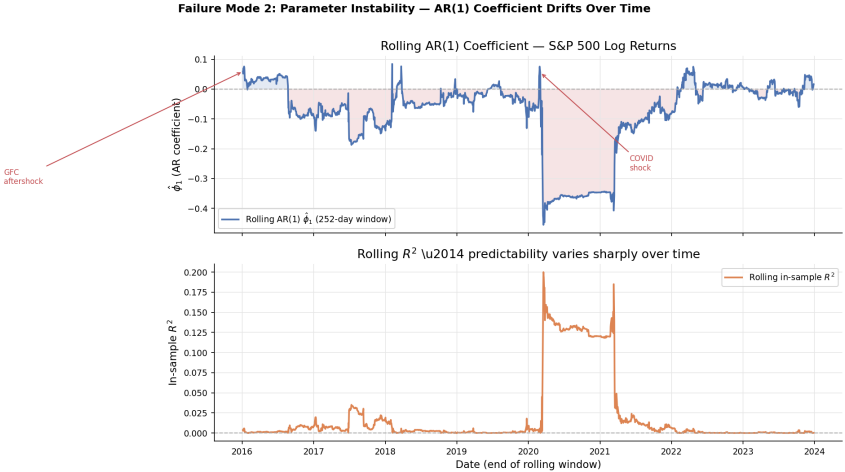
1. **Regime change**: the world shifts (policy, crisis, shock).
2. **Parameter drift**: relationships slowly move over time.
3. **Extrapolation**: future inputs leave historical support.

Key message: A model can be statistically correct and still operationally fragile.

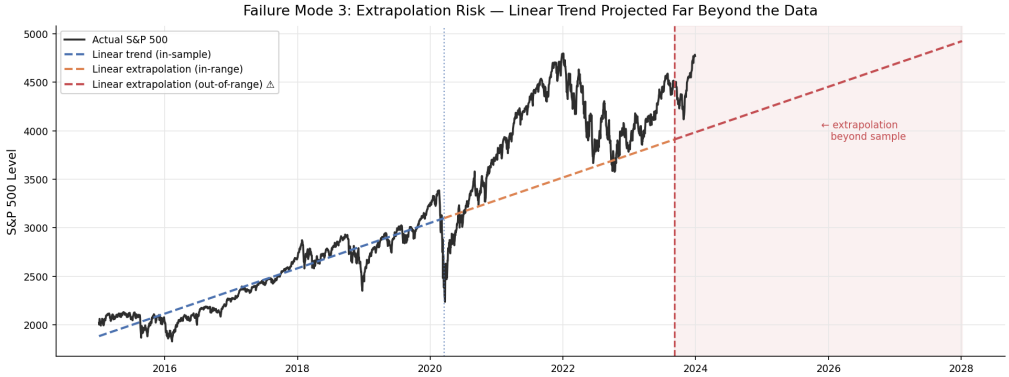
Failure Mode 1: Regime Change



Failure Mode 2: Parameter Instability



Failure Mode 3: Extrapolation Risk



Decision Checklist (Can I Trust This Forecast?)

Before deployment, ask:

1. Is the validation strictly time-aware?
2. Is there a large train-test performance gap?
3. Did we test robustness across windows/folds?
4. Do we have interval uncertainty, not only point forecasts?
5. What failure mode is most likely next period?

In-Class Discussion Prompts

- Which failure mode is most relevant in financial markets?
- If you must choose between lower RMSE and higher stability, what do you choose and why?
- What is one realistic leakage risk in a company forecasting pipeline?

Exit Ticket (3 Sentences)

1. Define overfitting in your own words.
2. Explain why random K-fold can be misleading for time series.
3. Name one reason a good model may fail next year.

End.